

Data Science

Lecture 09: Association Rule Mining

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Converted from LNKB-tex part-en-ds-chp7-association-rule-mining.tex. This is the probability chapter applied to counting: support is a joint probability, confidence a conditional one, and lift a test of independence.

Module 1

Itemsets and Their Search Space

***Guiding Question:** Ten thousand products give 2^{10000} candidate itemsets. How can this problem possibly be solvable?*

The Question

Given a set of transactions — shopping baskets, clicked pages, co-occurring symptoms — which **items appear together** more often than chance, and which **imply** which?

Support

$$P(A \cap B) > \sigma$$

How often do these items occur in the same transaction?

Confidence

$$P(A \mid B) > \gamma$$

Given B is in the transaction, how often is A too?

Both are quantities the students already know: support is a **joint** probability, confidence a **conditional** one, both estimated by counting transactions.

Say the connection explicitly — this whole chapter is the probability material applied to a counting problem. Nothing new is being defined, only named for a new context.

Formalisation

Let I be a set of **items** and X a set of **transactions**, each $x \subseteq I$.

- An **itemset** is any $A \subseteq I$ a **k -itemset** has $|A| = k$.
- Its **support** is $\sigma(A) = \frac{|\{x \in X : A \subseteq x\}|}{|X|}$.
- A is **frequent** if $\sigma(A) \geq \sigma_{\min}$.

The Search Space

All itemsets over I form a **lattice** ordered by inclusion — with $2^{|I|}$ elements.

A supermarket with 10,000 products has 2^{10000} candidate itemsets. Enumeration is impossible; the entire chapter is about avoiding it.

Support and confidence are joint and conditional probabilities estimated by counting. The difficulty is not the measure but the **size of the search space**.

The lattice picture matters for the next unit: A-Priori prunes whole subtrees of it, and the pruning rule only makes sense once the ordering is clear.

Module 2

The A-Priori Algorithm

Guiding Question: *What single structural fact about support lets us discard most of an exponential space without looking at the data?*

The Property That Makes It Possible

A-priori property (anti-monotonicity). Every subset of a frequent itemset is frequent — equivalently: if an itemset is **infrequent**, **every superset of it is infrequent too**.

Adding an item can only ever *reduce* the number of transactions containing the set. So an infrequent itemset prunes its entire superset subtree from the search.

The property is obvious once stated and is the whole algorithm. Ask students to justify it in one sentence before showing the consequence — "requiring more items cannot make a set more common".

The Algorithm

A-Priori

L^1 = all frequent 1-itemsets.

for $k = 2$ $L^{k-1} \neq \emptyset$ $k++$: **generate** candidates $C_k = \{a \cup b \mid a \in L^{k-1}, b \in L^1, b \not\subseteq a\}$
prune any candidate having an infrequent subset **count** the support of the survivors in one pass over the transactions $L^k = \{c \in C_k \mid \sigma(c) \geq \sigma_{\min}\}$

Generate

build k -candidates from frequent $(k-1)$ -itemsets

Prune

discard candidates with any infrequent subset — no data access needed

Count

one database pass per level

Cost and Improvements

What it costs

One pass over the transactions **per level k** , plus candidate generation. The bottleneck is I/O and the number of candidates.

How it is improved

Hashing candidates into buckets, transaction reduction (dropping transactions containing no frequent itemset), partitioning and sampling.

Mini-Exercise

$\{\text{bread, milk}\}$ has support 0.18, and $\sigma_{\min} = 0.22$.

1. Is $\{\text{bread, milk}\}$ frequent?
2. What can you say about $\{\text{bread, milk, eggs}\}$?
3. How many database passes did answering 2 require?

Mini-Exercise — Solution

A-Priori turns one structural fact — **infrequent sets have no frequent supersets** — into a level-wise search that prunes most of an exponential space without touching the data.

Mini-Exercise — Solution

1. No — $0.18 < 0.22$.

2. It is **infrequent**, necessarily: its support cannot exceed that of any subset.

3. **None.** The conclusion follows from the property alone. That is exactly where the savings come from.

A-Priori turns one structural fact — **infrequent sets have no frequent supersets** — into a level-wise search that prunes most of an exponential space without touching the data.

Module 3

From Itemsets to Rules

Guiding Question: *A rule holds 80% of the time — and tells you nothing. How is that possible, and what should you measure instead?*

Generating Rules

An **association rule** $A \Rightarrow B$ (with $A \cap B = \emptyset$) has

$$\text{support} = \sigma(A \cup B), \quad \text{confidence} = \frac{\sigma(A \cup B)}{\sigma(A)}$$

Every frequent itemset F is split in all possible ways into A and $B = F \setminus A$ rules meeting the confidence threshold are kept.

Support comes from the itemset and is already known. Only confidence must be computed — so rule generation needs **no further database passes**.

The two-stage structure (find frequent itemsets, then split them) is what makes the whole approach tractable. Confidence is a ratio of two numbers already counted.

Confidence Is Not Enough

Suppose 80% of all transactions contain milk. A rule $\text{bread} \Rightarrow \text{milk}$ with **confidence 0.80** has learned nothing: milk appears in 80% of baskets regardless of bread.

$$\text{lift}(A \Rightarrow B) = \frac{P(B \mid A)}{P(B)} = \frac{\sigma(A \cup B)}{\sigma(A)\sigma(B)}$$

lift > 1

A makes B **more** likely — a positive association.

lift = 1

A and B are **independent** — the rule carries no information.

lift < 1

A makes B **less** likely — the rule holds in reverse.

Lift is the independence test from Lecture 02 written as a ratio: it compares the joint probability against the product of the marginals.

This is the payoff of teaching independence properly. Confidence measures a conditional probability lift measures the *departure from independence*, which is what "association" was supposed to mean all along.

Mini-Exercise

A retailer reports: nappies $\Rightarrow$ beer, support 0.02, confidence 0.60. Beer appears in 55% of all transactions.

1. Compute the lift.
2. Is the rule actionable?
3. What would make it more convincing?

Mini-Exercise — Solution

Confidence alone rewards **popular** consequents. **Lift** corrects for the base rate by comparing against independence — the same correction Bayes' theorem makes for priors.

Close by connecting all three: support = joint, confidence = conditional, lift = departure from independence. Three probability concepts, one application — and the base-rate trap in a third guise.

Mini-Exercise — Solution

1. $\text{lift} = 0.60/0.55 \approx 1.09$ — barely above 1.

2. Hardly. A 9% relative increase over the base rate, on 2% of transactions, is a weak effect on a small slice.

3. A higher lift, a larger support, and evidence that it **persists** — across stores, seasons and time. And a mechanism: a rule with no explanation is a correlation.

Confidence alone rewards **popular** consequents. **Lift** corrects for the base rate by comparing against independence — the same correction Bayes' theorem makes for priors.